ONGC AI Knowledge Assistant

Enterprise Offline RAG Chatbot & Technical Architecture Report

Organization: Oil and Natural Gas Corporation (ONGC)	**Project Status:** Production Ready (v1.0.0)
Department: Policy Monitoring & Control (PMC)	**Security Level:** 100% Offline / Local Network
Division: Corporate Materials Management, New Delhi	**Date:** July 2026

1. Executive Summary

The **ONGC AI Knowledge Assistant** is an enterprise-grade, fully offline Artificial Intelligence chatbot engineered to query ONGC internal manuals, policy documents, and procurement guidelines securely and instantly. Built using a local **Retrieval-Augmented Generation (RAG)** pipeline, the system operates entirely within ONGC's isolated network environment without relying on external cloud APIs, eliminating third-party data privacy risks.

The system combines local LLMs via Ollama, ChromaDB vector storage, a FastAPI web backend, and an intuitive web dashboard featuring real-time token streaming, multi-session memory, PDF drag-and-drop ingestion, voice controls, and analytics.

Zero-Cloud Security Architecture: *All document parsing, embeddings generation, vector similarity searches, and LLM inferences execute locally on ONGC workstation hardware, complying fully with enterprise security policies.*

2. Core Features & Capabilities

- **Offline RAG Pipeline:** Answers user queries strictly grounded in indexed PDF manuals, eliminating AI hallucination.
- **Local LLM Flexibility:** Dynamic switching across local models including Llama 3.2 (1B/3B), Qwen 2.5, and Gemma 3 via Ollama.
- **Persistent Vector Storage:** ChromaDB database powered by sentence-transformers/all-MiniLM-L6-v2 dense embeddings.
- **Real-Time Token Streaming:** FastAPI Server-Sent Events (SSE) stream responses token-by-token for ultra-responsive interaction.
- **Multi-Session Web UI:** Browser localStorage session management with light/dark themes and responsive sidebar.
- **Document Ingestion Engine:** Drag-and-drop PDF parser with character chunking (500 size, 50 overlap) and auto-indexing.
- **User Feedback & Analytics:** SQLite rating database tracking thumbs up/down, user comments, and satisfaction metrics.
- **Voice Interface & Dockerization:** Native Web Speech STT/TTS voice integration and complete Docker Compose deployment container.

3. System Architecture & Technology Stack

Architecture Layer	Technology / Framework	Implementation Details & Responsibility
Frontend UI	HTML5, CSS3, Vanilla JS	Single-page app, Glassmorphism design system, Web Speech API, theme toggle
Backend API	FastAPI + Uvicorn	Async REST endpoints, streaming SSE response handler, static route mounting
RAG Orchestration	LangChain Framework	Prompt engineering, document context formatting, session history assembly
Local LLM Engine	Ollama API Host	Offline inference runtime hosting Llama 3.2, Qwen 2.5, and Gemma 3 models
Embeddings Model	all-MiniLM-L6-v2	384-dimensional dense sentence transformer running in offline mode
Vector Database	ChromaDB	Persistent similarity search vector database with HNSW indexing
Feedback DB	SQLite (feedback.db)	Stores conversation metrics, user ratings, and feedback comments
Deployment	Docker & Compose	Multi-stage container with persistent volume mounts for docs & vector DB

4. Key Software Modules

- **main.py** — FastAPI server exposing REST endpoints (/api/chat, /api/ingest, /api/documents, /api/analytics).
- **app/rag.py** — Core RAG logic supporting standard responses and SSE streaming token generation (run_rag_pipeline_stream).
- **utils/vectorstore.py** — ChromaDB CRUD store manager, similarity retriever factory, and document deduplication check.
- **utils/loader.py** — PDF text extractor utilizing PyPDFLoader and RecursiveCharacterTextSplitter (500 size, 50 overlap).
- **ingest.py** — CLI bulk PDF ingestion utility for populating ChromaDB from the docs/ repository in offline mode.
- **app/memory.py** — Session-based chat memory manager maintaining multi-turn context per user session ID.

5. Repository Directory & File Structure

```
ONGC_CHATBOT/
|-- app/
| |-- config.py # Paths, model defaults, hyperparameter settings
| |-- chatbot.py # Top-level chatbot wrapper
| |-- memory.py # Session conversation history memory manager
| +-- rag.py # Core RAG retrieval & SSE streaming pipeline
|-- utils/
| |-- embeddings.py # Offline HuggingFace embeddings singleton
| |-- loader.py # PDF document loader & text chunker
| +-- vectorstore.py # ChromaDB CRUD vector store operations
|-- prompts/
| +-- system_prompt.py # ONGC enterprise system prompt template
|-- templates/
| +-- index.html # Web dashboard HTML structure
|-- static/
| |-- style.css # Custom CSS styling (dark/light themes)
| +-- app.js # Frontend Javascript (chat, voice, ingestion)
|-- docs/ # PDF knowledge base directory
|-- chroma_db/ # Persistent ChromaDB vector database
|-- main.py # FastAPI application main server
|-- ingest.py # PDF document ingestion CLI tool
|-- Dockerfile # Container deployment script
+-- docker-compose.yml # Multi-container orchestration config
```

6. Operations & Deployment Guide

Option A: Standard Python Virtual Environment

1. Start Ollama local LLM service: ollama serve
2. Ingest PDF manuals into ChromaDB: python ingest.py
3. Launch web server: python -m uvicorn main:app --host 127.0.0.1 --port 8000
4. Open dashboard in web browser: http://127.0.0.1:8000

Option B: Docker Container Deployment

Run docker-compose up --build -d to launch containerized services. Persistent volumes ensure chroma_db/, docs/, and feedback.db data are preserved across container updates.

7. Future Roadmap & Strategic Enhancements

- **Hybrid Search (Dense + Sparse):** Integrate BM25 keyword matching alongside vector search for accurate clause number retrieval.
- **OCR & Multi-Modal Processing:** Incorporate PaddleOCR to index scanned image-based historical ONGC policy documents.
- **Role-Based Access Control (RBAC):** Enforce document-level access permissions based on employee department authorization levels.
- **Multilingual Capabilities:** Support seamless English to Hindi query transformation using dedicated Hindi fine-tuned checkpoints.

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